

The UNIVERSITY OF AZAD JAMMU & KASHMIR

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Domain Model

SUBMITTED BY:

STUDENT NAME:

SHAHNZA MUNIR

ROLL NO:

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INSTRUCTOR NAME:

Engr. Ghulam Rubani Butt

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Individual Short Report

Conceptual Classes and Abstraction in a University Course Registration System

1. Selected Real-World Domain

The selected domain is a University Course Registration System. It allows students to view available courses, register for courses, and check registration status. Administrators can manage course information and review student registrations. This is a familiar and manageable domain because it represents common activities performed by students and university staff.

2. Candidate Conceptual Classes

No.	Conceptual Class	Justification
1	Student	Represents a student who uses the system to browse and register for courses.
2	Course	Stores the essential information about a course offered by the university.
3	Instructor	Represents the teacher responsible for teaching a course.
4	Administrator	Represents an authorized staff member who manages courses and registrations.
5	CourseRegistration	Records a student's registration for a particular course and its status.
6	Department	Groups related courses and instructors under an academic department.
7	Semester	Identifies the academic term in which courses are offered.
8	Classroom	Represents the room or location assigned for a course.
9	Timetable	Stores the scheduled days and times for course sessions.
10	Prerequisite	Represents a course or requirement that must be completed before another course.

3. Attributes and Methods for Three Classes

A. Student

Attributes: studentId, fullName, email, program, semester

Methods: viewAvailableCourses(), registerForCourse(), viewMyRegistrations(), cancelRegistration()

B. Course

Attributes: courseCode, courseName, creditHours, instructorName, description

Methods: displayDetails(), checkAvailability(), updateCourseInformation()

C. CourseRegistration

Attributes: registrationId, studentId, courseCode, registrationDate, status

Methods: submitRegistration(), approveRegistration(), rejectRegistration(), getStatus()

4. Main Abstraction Decisions

The model includes only information needed to support course browsing, registration, and registration management. For example, Student includes an ID, name, email, program, and semester, but does not include personal details such as home address, family information, or hobbies because these details are not required for course registration. Course focuses on its code, name, credit hours, instructor, and description rather than classroom equipment or the instructor's personal life. The CourseRegistration class is kept separate from Student and Course because one student can register for several courses and each course can have registrations from many students. The model also avoids implementation details such as database table names, page colours, and button designs; these belong to the software's implementation or user interface, not its conceptual model.

5. One Invented Software Object

Object name: RegistrationEligibilityChecker

RegistrationEligibilityChecker is a proposed software object that checks whether a student is allowed to register for a selected course. It could verify that the student has not already registered, that the course has available seats, and that required prerequisites have been completed. It would return a clear result explaining whether registration can proceed. This object would help keep eligibility rules in one place and reduce duplicated checking in different parts of the system.

Conclusion

A clear conceptual model helps developers understand the important entities in a university course registration system before implementation. Identifying classes, their attributes and methods, and the information intentionally excluded provides a focused design that is easier to develop, test, and maintain.